

tions. It also provides authentication and authorisation for secure communication between sensors. It helps organisations maintain end-to-end supply chain integrity using enterprise-grade encryption technologies and management capabilities.

- IoT device manufacturing and maintenance can be done through Android app development for IoT applications, such as an IoT-connected solution for industry automation and connected vehicles.

Rajkiran, director, Thingworx OEM Business, says, "In the future, solution providers who use application-development platforms will be in a commanding position to drive their businesses. If they can create the right application for the right industry, they will be able to work on that use case and update the application for other purposes in a matter of hours."

Business opportunities in the development of smart cities and smart buildings

The IoT is transforming smart buildings to develop smart cities at a rapid pace. Organisations are seeking opportunities to enter new markets, develop new products and drive differentiated customer experiences with the IoT.

Smart cities and smart buildings have several business opportunities in data monetisation, connected products and experiences, smart monitoring, asset management and optimisation. Remote structure monitoring from anywhere in addition to re-use of sensors on multiple locations help build better buildings. A connected home makes devices such as coffee pots, appliances and security cameras more useful to improve a consumer's life.

A smart building is comparatively easier to design and maintain as compared to a smart city. A smart city needs more cost-efficient municipal services, enhanced public transportation, good health facilities and, most importantly, a secure place to live and work. Smart city technology generates a significant amount of valuable data that need to be harnessed, analysed and acted upon.

Use of the IoT in the energy sector improves production, reduces cost, increases worker safety and improves

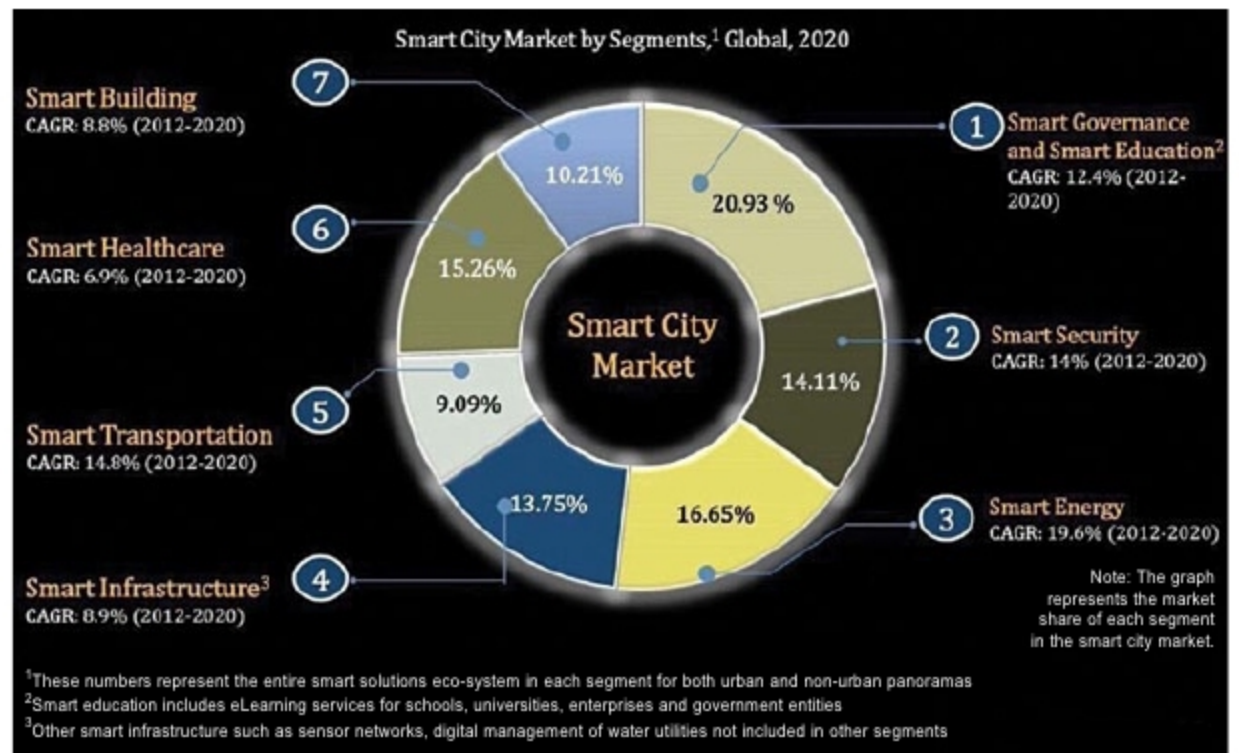


Fig. 3: Smart cities to create huge business opportunities (Credit: Frost & Sullivan)

electrical efficiency with IoT energy solutions. Smart grids and meters monitor and control unnecessary usage of electrical systems. This has made possible remote data management and monitoring, automation and control, systems for effective transmission and distribution networks.

Energy bills based on actual consumption improve the billing process and contribute to improved cash flow. Smart metering is also an enabler for new prepaid energy services that can be offered to customers who lack the funds for an up-front deposit or high pre-payments.

Easily-connected health devices remotely monitor and manage patients, and get real-time analysis of data.

IoT devices help get real-time visibility for road conditions and preventative maintenance capabilities for connected cars, roadways and transit systems.

These also help employers ensure worker safety and enhance overall workplace security. Embedded sensors and wearables continuously monitor workers in high-risk environments, like mining, heavy industries and construction, and warn against falls and exhaustion.

Connected video surveillance cameras monitor office premises and ensure protection of important assets.

The IoT enables users to connect different edge devices with each other and operate on them seamlessly, from

a remote location with IoT solutions.

Smart irrigation systems and digital dairy practices in rural areas have further increased growth opportunities.

Retailers can manage inventory in real time. Sensors and IoT mobile applications offer market insights that elevate customer experience, enhance convenience levels, and influence online and in-store behaviours.

Choose the right IoT solutions for your business

Every sector requires the power of the latest technologies to easily resolve complex problems and deliver simpler solutions. From tracking the flow of people and traffic density to monitoring fluctuations in office temperature, crop moisture levels or vital signs of patients, solutions are needed to manage over multiple data points. Selecting the wrong components, software or network technology can make IoT projects less likely to be successful, more expensive, and harder to manage and sell.

With careful strategy and planning, adoption of IoT solutions can result in large benefits for businesses by leveraging quick processing, storage and retrieval of data. The IoT can provide a wide range of improvements, from ensuring better control over production and costs to development of new offers. It takes businesses to the next level by enabling more proactive decision-making. **EFY**